



"If 386BSD had been available when I started on Linux, Linux would probably never have happened"—Linus Torvalds

## China

Is developing a secure OS for use in it's government sytems, based on FreeBSD

appropriate key combination. The beauty of multiple terminals is that one can issue a command in one, and not wait for the routine to end to execute another command, the latter can be entered at another virtual terminal and will be executed parallelly.

Get to a virtual terminal and log in as root and issue the `dmesg` command. This will output a flurry of system messages most of which will not be readable. So issuing a `dmesg | more` command is better since this will load a page of information and wait for the user to press the Space button before loading the next page. If you know the on-board audio chip name, then you can also search for the specific name with `dmesg "Intel"` or similar. This will only display those messages containing the text.

The next step is to load a generic sound driver (`snd_driver`) and check if the hardware gets detected. To load the driver issue the `kldload snd_driver` command. This will generate a list of entries if the hardware gets detected, and rerunning the `dmesg` command will show the details of the newly detected hardware. You can test if the audio system is working by playing a music file. To ensure that the appropriate driver is loaded at boot time the `/boot/loader.conf` file needs to be edited. Issue a `ee /boot/loader.conf` command and add the line in the following format `<sound_driver>_load="YES"`.

## Application Management

FB offers two ways to get additional apps installed in the system. One is as Package and the other is as Port.

### Package Management

Package refers to pre-compiled application executable/binary file. Since it is pre-compiled, the installation process is much faster. To install a package, the best way is to rely on the `sysinstall` tool which was also used for the first installation. `sysinstall` uses the command `pkg_add` in the background



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to do the actual installation, so one can also directly install an application by this command, bypassing `sysinstall`. But the menu driven `sysinstall` has its advantages.

To install applications from the installation media, first insert the media in the drive, and log in as root at any virtual terminal. There issue the `sysinstall` command. The familiar `sysinstall` interface pops up. Here select the **Post-installation Configuration** menu, and in that menu select the **Install Pre-packaged software for FB** option. After the source is specified, the list of apps in the source are listed. We shall use the installation media to install Opera, so the source is **FreeBSD CD/DVD**.

Opera 9.6 is present under the **www** menu. Once the entry is selected, and the install confirmed, the required files are extracted and placed in the appropriate folders.

### Ports collection.

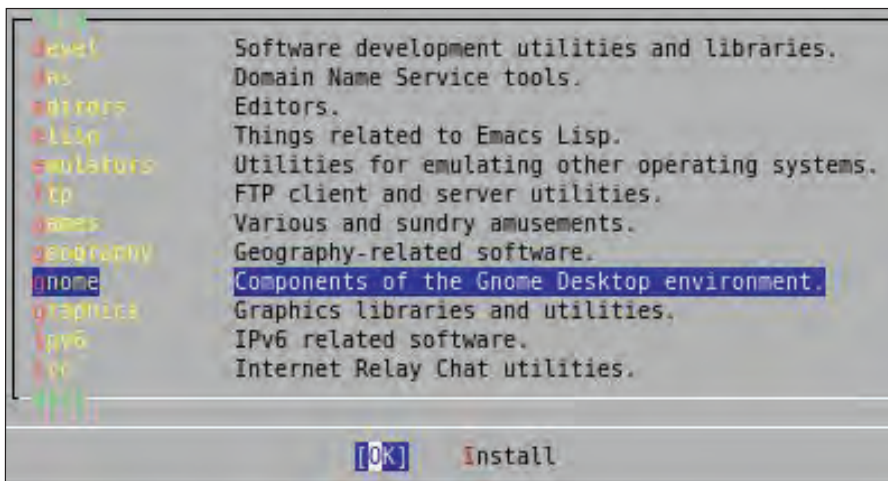
One way to get apps is to search the Ports tree. The **Ports Collection** system is a unique feature of BSD, and the **Ports** tree is at the centre of a system of installing applications after compiling them from source. During the OS installation, the user permission is sought to install the Ports collection, which gets stored under the `/usr/ports` path. There are thousands of folders in

the Ports directory, with each folder containing information regarding an application of the same name. Please note, that the folder DOES NOT contain the source of the application, merely the information that would be needed to successfully compile and install the application when the need arises.

Unfortunately, there is no interface to meaningfully browse the Ports collection. The closest one can get is by downloading a listing of the Ports collection as HTML file from the `freebsd.org` site and searching for an application or activity by name. If there are matches, read the application description to verify if it fits the need. If an application name is known, then the search is much easier -from within KDE, the `/usr/ports` folder can be searched. It is important to note the path of the folder matching the application name.

## Conclusion

At the end of the journey, it would be fair to state that the BSD membership doesn't come cheap and needs to be earned! The fact that BSD focussed on server based installations where GUIs are not welcome and automounting is a security threat explains the state of affairs to some extent. Newer distros that are better suited for desktop use and come in live CD format (which is considerably more attractive than slicing up the hard disk) can change this scenario. 



Pre-compiled packages included with distro